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The no-wait flow shop with rejection

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We consider the no-wait flow shop scheduling problem with rejection. We first show how to augment the travelling salesman problem formulation for the no-wait flow shop minimum makespan problem to incorporate the rejection option. We then focus on polynomially solvable cases by considering problems with ordered jobs. We present a third-order polynomial-time dynamic programming algorithm to minimise the sum of makespan and total rejection cost and faster quadratic algorithms for two special cases. We also present a third-order polynomial-time dynamic programming algorithm to minimise the sum of total completion time and total rejection cost with ordered jobs. We also exploit the duality between no-wait and no-idle flow shops to determine which operations to outsource in a no-idle flow shop with an outsourcing option.

Keywords: scheduling; flow shop; no-wait; no-idle; rejection

1. Introduction

The bulk of the traditional scheduling literature assumes that all jobs must be scheduled. A stream of scheduling research surveyed by Shabtay, Gaspar, and Kaspi (2013) considers the option of rejecting jobs from the schedule. In that case, management has the option of rejecting a job that does not fit the current work plan incurring a rejection penalty. It is usually assumed that the rejected jobs are outsourced and the rejection penalty represents the outsourcing cost. Even though rejection and outsourcing are structurally similar, there are conceptual differences between them.

In scheduling problems with rejection, jobs are partitioned into scheduled and rejected respectively and the scheduled jobs are sequenced so that a composite objective function comprising the sum of a scheduling objective and total rejection cost is minimised. The most commonly used objective functions are the sum of makespan of scheduled jobs and total rejection cost or the sum of total completion time of scheduled jobs and total rejection cost.

A processing mode that has not been considered in the literature in conjunction with rejection is the no-wait restriction in flow shops (see the recent survey of Allahverdi (2016) on no-wait shops where around 300 papers are reviewed without a single one considering the rejection option). Recent results for multi-criteria no-wait flow shop scheduling are presented by Ye, Li, and Nault (2019). In a no-wait flow shop, the start time of each operation must coincide with the finish time of the previous one for all jobs; equivalently, a job is processed from start to finish without interruptions. The no-wait restriction is imposed either to prevent degradation of the product or because of the prohibitively high cost of customer waiting in certain service industries.

The objective of this paper is to analyse the no-wait flow shop with the rejection option. We first show how to augment the travelling salesman problem formulation for the no-wait flow shop minimum makespan problem to incorporate the rejection option. We then focus on polynomially solvable cases by considering problems with ordered jobs defined in the next section. We present a forward $O(n^3)$ dynamic programming (DP) algorithm to minimise the sum of makespan and total rejection cost and faster quadratic and linear time algorithms for two special cases. We also present a backward $O(n^3)$ DP algorithm to minimise the sum of total completion time and total rejection cost with ordered jobs.

Our results can also be used in a no-idle flow shop in which each machine must be kept continuously busy after it starts processing (see Goncharov and Sevastyanov (2009) for a survey of this model). If only permutation schedules are allowed, Kalczyński and Kamburowski (2007) observed a duality between no-idle and no-wait flow shops according to which jobs and machines can be transposed. We exploit this duality to determine which operations to outsource in a no-idle flow shop with an outsourcing option. To the best of our knowledge, our results for the no-idle flow shop are the only results in the flow shop literature focused on outsourcing entire processes rather than entire jobs. By utilising our results, management has the option of outsourcing an entire operation in a serial assembly line by comparing the outsourcing cost against the realised

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benefit from speeding up the line due to processing in-house fewer operations. Previous research on partial outsourcing focused on investigating the outsourcing of certain operations for certain jobs and not the outsourcing of an entire operation for all jobs; see for example, the papers by Lee and Choi (2011), Choi and Chung (2011) and Gao and Lu (2014) among others.

The rest of the paper is organised as follows: In the next section, we analyse the no-wait flow shop minimum makespan problem with the rejection option. The same problem with ordered jobs is analysed in Section 3. Some special cases of the problem with ordered jobs are analysed in Section 4. The no-wait flow shop with rejection, ordered jobs and the total completion time objective is analysed in Section 5. The conclusions of this research are summarised in Section 6.

2. The no-wait flow shop with rejection and the makespan objective

The m -machine no-wait flow shop $Fm|nwt|C_{\max}$ problem (with arbitrary job processing times, a fixed number of machines (m) and the minimum makespan objective) is solvable in $O(n \log n)$ time by the algorithm of Gilmore and Gomory (1964) when $m = 2$ and each job must visit both machines; the $Fm|nwt|C_{\max}$ problem is strongly NP-hard when $m \geq 3$ (Rock 1984a). In a no-wait flow shop, permutation schedules (in which the jobs are processed in the same order on all machines) are the only feasible schedules because a job cannot ‘pass’ another job in a sequence.

Let p_{ij} denote the processing time of job $j = 1, \dots, n$ on machine M_i , $i = 1, \dots, m$ and C_{ij} denote the completion time of job j on machine M_i . The no-wait restriction necessitates the uninterrupted processing of each job; therefore, $C_{ij} = C_{i-1,j} + p_{ij}$ for $i = 2, \dots, m$. As a result, when job j immediately follows job k , the increase in the makespan d_{kj} can be determined using only the processing times of jobs j and k ; Emmons and Mathur (1995) showed that

$$d_{kj} = \max_{r=1, \dots, m} \left\{ p_{rk} + \sum_{i=r}^m (p_{ij} - p_{ik}) \right\}. \quad (1)$$

This facilitates the well-known approach of formulating the $Fm|nwt|C_{\max}$ problem as a Travelling Salesman Problem (TSP) by viewing the tour of the travelling salesman as the cyclic permutation of the n jobs; define the binary variables

$$x_{jl} = \begin{cases} 1 & \text{if job } j \text{ is in position } l \\ 0 & \text{otherwise} \end{cases} \quad \text{for } j = 1, \dots, n; l = 1, \dots, n.$$

Then, the TSP formulation is:

$$\begin{aligned} \min & \sum_{l=2}^n \sum_{k,j} c_{kj} x_{k,l-1} x_{jl} \\ \text{s.t.} & \sum_{l=1}^n x_{jl} = 1, \quad j = 1, \dots, n \\ & \sum_{j=1}^n x_{jl} = 1, \quad l = 1, \dots, n, \quad x_{jl} \in \{0, 1\} \end{aligned}$$

where $c_{kj} = d_{kj}$ is given by (1) for $k = 1, \dots, n$; $j = 1, \dots, n$; $k \neq j$ and $c_{jj} = \sum_{i=1}^m p_{ij}$ for all $j = 1, \dots, n$. The objective function determines the makespan of the $Fm|nwt|C_{\max}$ problem and not the full length of the cyclic tour of the travelling salesman. The latter is determined by adding the term $\sum_{k,j} c_{kj} x_{k,n} x_{j1}$ to the objective function. The flow shop processing order is preserved because the makespan increment of $c_{kj} = d_{kj}$ is actually added to the makespan only when jobs k, j occupy adjacent positions in the permutation sequence because of the presence of the quadratic terms $x_{k,l-1} x_{jl}$.

The above formulation is actually a quadratic assignment problem formulation (Burkard, Dell’Amico, and Martello 2009, 210). It is stated there that problems with up to 30 jobs can be solved by commercially available optimisation packages. Our motivation for considering this formulation is to show that it can be augmented to incorporate the job rejection option as detailed next.

In the $Fm|nwt|C_{\max}$ problem with rejection, management has the option of rejecting a job j from the set of scheduled jobs in S by paying its rejection penalty e_j ; the problem is denoted as $Fm|nwt|C_{\max} + \sum e_j$ where $C_{\max} = \max_{j \in S} \{C_j\}$ is the makespan of the scheduled jobs in S and $\sum_{j \in \bar{S}} e_j$ is the total rejection cost for the complementary set of the rejected jobs in $\bar{S}$.

A no-wait scheduling problem with rejection is at least as hard as its counterpart without rejection. Therefore, the $Fm|nwt|C_{\max} + \sum e_j$ problem is strongly NP-hard when $m \geq 3$ and its complexity is open when $m = 2$.

Our first result is to show that the $Fm|nwt|C_{\max} + \sum e_j$ problem can be formulated as the following augmented $(2n \times 2n)$ TSP:

$$\begin{aligned} \min & \sum_{l=2}^{2n} \sum_{k,j} c_{kj} x_{k,l-1} x_{jl} \\ \text{s.t.} & \sum_{l=1}^{2n} x_{jl} = 1, \quad j = 1, \dots, 2n \\ & \sum_{j=1}^{2n} x_{jl} = 1, \quad l = 1, \dots, 2n, \quad x_{jl} = \{0, 1\} \end{aligned}$$

$c_{kj} = d_{kj}$ where d_{kj} is given by (1) for $k = 1, \dots, n; j = 1, \dots, n; k \neq j$;

$c_{jj} = \sum_{i=1}^m p_{ij}$ for all $j = 1, \dots, n$;

$c_{kj} = \infty$ for $k = 1, \dots, n; j = n+1, \dots, 2n$;

$c_{kj} = \infty$ for $k = n+1, \dots, 2n; j = 1, \dots, n$;

$c_{kj} = \infty$ for $k = n+1, \dots, 2n; j = n+1, \dots, 2n; k \neq j$;

$c_{jj} = e_j$ for all $j = n+1, \dots, 2n$;

In the above formulation, the additional positions $n+1, \dots, 2n$ in the sequence correspond to the option of rejecting job $j = 1, \dots, n$. The assignment constraints ensure that a job is either scheduled or rejected while all c_{kj} adjacency costs between a scheduled and a rejected job or between two rejected jobs are set to infinity. Also, any rejected job j can only be paired with itself incurring the cost of $c_{jj} = e_j$.

3. The $Fm|nwt|C_{\max} + \sum e_j$ problem with ordered jobs

The complexity of the $Fm|nwt|C_{\max} + \sum e_j$ problem when $m \geq 3$ motivates the consideration of flow shops with specially-structured job processing times so that the $Fm|nwt|C_{\max} + \sum e_j$ problem can be solved in polynomial time. One such shop is the ordered flow shop introduced by Smith, Panwalkar, and Dudek (1975); in an ordered shop, the following conditions hold.

If $p_{ij} < p_{ik}$ for any two jobs j, k on any machine M_i , then $p_{ij} \leq p_{ik}$ on all machines $M_i, i = 1, \dots, m$ (ordered jobs) (2a);
if $p_{ij} < p_{kj}$ for any job j on any two machines M_i, M_k , then $p_{ij} \leq p_{kj}$ for all jobs $j, j = 1, \dots, n$ (ordered machines) (2b).

Condition (2a) holds whenever $p_{ij} \leq p_{ik}$ on all machines for any two jobs j, k implying that jobs can be ordered from smallest to largest. More formally, if only (2a) holds, we have a semi-ordered flow shop with ordered jobs; we denote this problem by inserting the term *sorj* in the problem definition. In that case, the shortest processing (SPT) and longest processing time (LPT) sequences are well defined and identical on all machines.

Panwalkar, Smith, and Koulamas (2013) reviewed ordered and semi-ordered flow shop problems and stated that ordered problems can be related to practical situations found in many industries. Specifically, an ordering of processing times according to jobs leading to ordered jobs can be caused by several factors such as the number of units or parts of a job. If the nature of the operations of each job is very similar but the jobs differ greatly in the number of units of production, then the processing times of the jobs satisfy the ordered jobs assumption.

There is continued interest in ordered flow shop and proportionate flow shop scheduling research following the survey of Panwalkar, Smith, and Koulamas (2013) including the paper by Lee, Zheng, and Pinedo (2019) focused on online ordered flow shops and the paper by Kovalev et al. (2019) focused on no-wait proportionate flow shops among others. The reader interested in earlier ordered flow shop papers is directed to the survey of Panwalkar, Smith, and Koulamas (2013).

We show next that the $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem can be solved in $O(n^3)$ time by dynamic programming. According to Arora and Rana (1980), the scheduled jobs in an optimal solution for the $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem should be sequenced in a pyramidal (SPT-LPT) sequence so that all jobs preceding the longest scheduled job are sequenced in the SPT order and all jobs following the longest scheduled job are sequenced in the LPT order.

Let jobs in an instance of the $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem be indexed in the LPT order. Let state $S(j, l, r)$ define an optimal solution for the job subset $\{1, \dots, j\}$ subject to the condition that the scheduled jobs among $\{1, \dots, j\}$ are processed

without any intentionally inserted idle time (that is, at least one machine is processing a job at any point in time), l is the leftmost scheduled job ($1 \leq l \leq j$) and r is the rightmost scheduled job ($1 \leq r \leq j$); state $S(j, 0, 0)$ is defined as the state where all jobs $\{1, \dots, j\}$ have been rejected.

If job j is appended to the right of a partial schedule ending with job k , the increase in the makespan d_{kj} is given by (1). If there is no job scheduled prior to j , then $k = 0$ and $d_{0j} = \sum_{i=1}^m p_{ij}$. By symmetry, if j immediately precedes job k , then the increase in the makespan d'_{jk} is given by $d'_{jk} = \max_{r=1, \dots, m} \{p_{rk} + \sum_{i=1}^r (p_{ij} - p_{ik})\}$ or by $d'_{j0} = d_{0j} = \sum_{i=1}^m p_{ij}$ if $k = 0$.

Let $F_j(l, r)$ denote the optimal solution value for state $S(j, l, r)$. The initialisation is

$$F_j(l, r) = \begin{cases} 0 & \text{if } j = l = r = 0 \\ \infty & \text{otherwise} \end{cases}$$

and the recursion for $j = 1, \dots, n, l = 0, \dots, j, r = 0, \dots, j$ is

$$F_j(l, r) = \begin{cases} F_{j-1}(0, 0) + e_j; & l = r = 0 & (3.1) \\ F_{j-1}(l, r) + e_j; & 1 \leq l < j, 1 \leq r < j & (3.2) \\ F_{j-1}(0, 0) + d_{0j}; & l = r = j & (3.3) \\ \min_{1 \leq r' \leq j-1} [F_{j-1}(l, r') + d_{rj}]; & 1 \leq l < j, r = j & (3.4) \\ \min_{1 \leq l' \leq j-1} [F_{j-1}(l', r) + d'_{jl'}]; & l = j, 1 \leq r < j & (3.5) \end{cases}$$

The optimal solution value is $F^* = \min_{0 \leq l \leq n, 0 \leq r \leq n} F_n(l, r)$. The recursive Equations (3.1)–(3.5) generalise the recursive equation of Axsater (1982) and partition the jobs optimally into scheduled and rejected respectively.

LEMMA 1 *The DP algorithm (3) supplies an optimal solution for the $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem.*

Proof of Lemma 1 The initial LPT order ensures that all partial schedules are SPT-LPT schedules. The justification for the recursive Equations (3.1)–(3.5) is as follows. There are three options while considering job j ; it can be either rejected or appended to the left (right) of the partial schedule. If $l < j$ and $r < j$, then j is rejected and $F_j(l, r)$ is obtained from $F_{j-1}(l, r)$ increased by e_j using (3.1) when all jobs $\{1, \dots, j\}$ have been rejected and (3.2) when at least one of jobs $\{1, \dots, j\}$ has been scheduled. If j is the only scheduled job, then $F_j(l, r)$ is computed using (3.3). If j is scheduled as the rightmost job, then $F_j(l, r)$ is computed using (3.4) by considering all $F_{j-1}(l, r')$ states (where r' is the rightmost job scheduled prior to considering j) in order to choose the option with the minimum increase in the objective function value. If j is scheduled as the leftmost job, then $F_j(l, r)$ is computed by (3.5) using a similar approach as in (3.4). ■

LEMMA 2 *The DP algorithm (3) runs in $O(n^3)$ time.*

Proof of Lemma 2 When $l < j$ and $r < j$, there are at most $O(n^3)$ states each computed in constant time; when either $l = j$ or $r = j$, there are at most $O(n^2)$ states computed in $O(n)$ time each. Therefore, the DP algorithm runs in $O(n^3)$ time. ■

PROPOSITION 1 *The $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem is solvable in $O(n^3)$ time.*

Proof of Proposition 1 It follows directly from Lemmas 1 and 2. ■

The third order $O(n^3)$ computational requirements of the DP algorithm (3) stems from the three options while adding a new job to the DP formulation; this job can be either rejected or appended to the left or appended to the right of the partial schedule. If job rejection is not an option, a newly added job to the DP formulation should be appended either to the left or to the right of the partial schedule; in that case, the computational requirements of the DP algorithm should be $O(n^2)$ reduced by an order of magnitude. This is indeed the case as shown by Axsater (1982).

If the no-wait restriction is lifted, the resulting $Fm|sorj|C_{\max} + \sum e_j$ problem is NP-hard because Choi and Chung (2011) showed that the more specialised $F2|ord|C_{\max} + \sum e_j$ problem (with both ordered jobs and ordered machines) is ordinary NP-hard. In that case, the contribution of an accepted job to the makespan does not depend only on the preceding job; instead, it depends on the machine ready times which leads to the $O(nP_1P_2)$ pseudo-polynomial DP algorithm for the $F2||C_{\max} + \sum e_j$ problem proposed by Choi and Chung (2011) where $P_1 = \sum_{j=1}^n p_{1j}$ and $P_2 = \sum_{j=1}^n p_{2j}$.

Shabtay and Gaspar (2012) proposed a faster $O(nP_2)$ DP algorithm for the $F2||C_{\max} + \sum e_j$ problem by utilising the difference of the machine ready times in the state definition. It is possible that multiple states with the same difference of

machine ready times have the same objective function value $TC = \max_{k \in S} \{C_k\} + \sum_{k \in \bar{S}} e_k$ because more than one pair of $\max_{k \in S} \{C_k\}$, $\sum_{k \in \bar{S}} e_k$ values may lead to the same TC value; in that case, all states with the same difference of machine ready times must be stored which implies that the worst-case running time of the DP algorithm of Shabtay and Gaspar (2012) is actually $O(nP_1P_2)$.

4. Special cases of the no-wait flow shop with rejection, ordered jobs and the makespan objective

When each task has its largest processing time on the first (last) machine, there is a maximal first (last) machine. Arora and Rana (1980) showed that the LPT (SPT) sequence is optimal for the $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem with a maximal first (last) machine. The optimal schedule for the corresponding problem with rejection can be built by appending jobs only to the right of a partial schedule leading to a faster $O(n^2)$ DP algorithm outlined below with a maximal last machine.

Let jobs be indexed in the SPT order. Define state $S(j, r)$ as the $S(j, l, r)$ state of the previous section when new jobs are only appended to the right of a partial schedule. Let $F_j(r)$ denote the optimal solution value of any schedule in state $S(j, r)$. The initialisation is

$$F_j(r) = \begin{cases} 0 & \text{if } j = r = 0 \\ \infty & \text{otherwise} \end{cases}$$

and the recursion for $j = 1, \dots, n, r = 0, \dots, j$ is

$$F_j(r) = \begin{cases} F_{j-1}(r) + e_j & \text{if } r < j \\ \min_{0 \leq r' \leq j-1} \{F_{j-1}(r') + d_{r'j}\} & \text{if } r = j \end{cases}$$

The optimal solution value is $F^* = \min_{0 \leq r \leq n} F_n(r)$. The following proposition can now be stated; its proof is similar to the proof of Proposition 1.

PROPOSITION 2 *The $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem with a maximal first (last) machine is solvable in $O(n^2)$ time. ■*

Proposition 2 holds for the more specialised fully-ordered $Fm|nwt, ord|C_{\max} + \sum e_j$ problem with $M_1 \leq M_2 \leq \dots \leq M_m$. By the duality between no-wait and no-idle flow shops, a solution to the above problem is also a solution to the no-idle flow shop $Fm|nid, ord|C_{\max} + \sum e_j$ problem with $M_1 \leq M_2 \leq \dots \leq M_m$ and the rejection costs corresponding to operations (machines) rather than corresponding to jobs. In the no-idle flow shop problem, management has the option of outsourcing a certain operation $M_i, i = 1, \dots, m$ for all jobs by paying the operation (machine) outsourcing cost e_i , effectively reducing the number of processing stages by one. The objective is to minimise the sum of makespan and total outsourcing cost; the optimal solution for the $Fm|nid, ord|C_{\max} + \sum e_j$ problem with $M_1 \leq M_2 \leq \dots \leq M_m$ can be determined according to Proposition 2.

Another special case is when all jobs have the same processing time on one machine. In that case, there is no maximal first or last machine. We outline the solution procedure for the $F2|nwt, p_{2j} = p|C_{\max} + \sum e_j$ problem. The job with the minimal p_{1j} value among all scheduled jobs is sequenced first in an optimal sequence and the remaining jobs are sequenced in any order. The contribution of any scheduled job j to the makespan (other than the first job in the sequence) is equal to $\max\{p_{1j}, p\}$ and is independent of its position in the sequence and also independent of the number of accepted jobs. Therefore, if the first job in the sequence is known, any other job j should be scheduled if $\max\{p_{1j}, p\} < e_j$ and should be rejected otherwise. These observations motivate the following optimal Algorithm A1 for the $F2|nwt, p_{2j} = p|C_{\max} + \sum e_j$ problem.

Algorithm A1

Step 1: For $j = 1, \dots, n$

 For $k = 1, \dots, j-1, j+1, \dots, n$

 If $\max\{p_{1k}, p\} < e_k$, then $k \in S$, else $k \in \bar{S}$

 Next k .

 Let $TC_j = p_{1j} + p + \max_{k \in S, k \neq j} \{p_{1k}, p\} + \sum_{k \in \bar{S}} e_k$

 Next j .

Step 2: $TC^* = \min_{j=1, \dots, n} \{TC_j\}$

End.

PROPOSITION 3 *The $F2|nwt, p_{2j} = p|C_{\max} + \sum e_j$ problem and the symmetric $F2|nwt, p_{1j} = p|C_{\max} + \sum e_j$ problem are solvable in $O(n^2)$ time by Algorithm A1.*

Proof of Proposition 3 There are n options for the job occupying the first position in an optimal sequence evaluated in the outer loop of Step 1. For each such option, the inner loop of Step 1 considers all remaining jobs one by one and either sequences them or rejects them optimally; therefore A1 is optimal. The inner loop of Step 1 is executed in constant time; therefore, A1 runs in $O(n^2)$ time. ■

5. The no-wait flow shop with rejection, ordered jobs and the total completion time objective

Let us consider the $Fm|nwt| \sum C_j + \sum e_j$ problem with the objective of minimising the sum of job completion times for the scheduled jobs in S added to the sum of rejection penalties for the rejected jobs in $\bar{S}$. The $Fm|nwt| \sum C_j + \sum e_j$ problem is strongly NP-hard because the corresponding $Fm|nwt| \sum C_j$ problem (without job rejection) is strongly NP-hard (Rock 1984b).

The $Fm|nwt| \sum C_j + \sum e_j$ problem can be formulated as an augmented $(2n \times 2n)$ TSP similar to its makespan counterpart by replacing its the objective function with $\min \sum_{l=2}^n (n-l+1) \sum_{k,j} c_{kj} x_{k,l-1} x_{jl}$. In this objective function, when job j is sequenced in position l immediately preceded by job k , the increment to the makespan equal to $\sum_{k,j} c_{kj} x_{k,l-1} x_{jl}$ is added to the completion times of all jobs sequenced in positions $l, \dots, n$ in agreement with the total job completion time objective.

In the remaining of this section, we focus on polynomially solvable cases of the $Fm|nwt| \sum C_j + \sum e_j$ problem with ordered jobs. van der Veen and van Dal (1991) showed that the SPT sequence is optimal for the $Fm|nwt, sorj| \sum C_j$ problem. This facilitates the solution of the $Fm|nwt, sorj| \sum C_j + \sum e_j$ problem in $O(n^3)$ time by implementing the following backward DP algorithm.

Let jobs be re-indexed in the SPT order. Let state $S(j, k, l)$ define an optimal solution for jobs $\{j, \dots, n\}$ subject to the condition that k jobs among them are scheduled ($k \leq n-j+1$), at least one machine is processing a job at any point in time and l is the leftmost scheduled job ($j \leq l \leq n$). Let $F_j(k, l)$ denote the optimal solution value for $S(j, k, l)$. The initialisation is

$$F_j(k, l) = \begin{cases} 0 & \text{if } j = n+1, k = 0, l = n+1 \\ \infty & \text{otherwise} \end{cases}$$

and the recursion for $j = n, \dots, 1, k = 0, \dots, n-j+1, l = j, \dots, n$ is

$$F_j(k, l) = \begin{cases} F_{j+1}(k, l) + e_j & \text{if } l > j \\ \min_{j+1 \leq l' \leq n} \{F_{j+1}(k-1, l') + \sum_{i=1}^m p_{ij} + (k-1)d'_{jl'}\} & \text{if } l = j \end{cases} \quad (4)$$

The optimal solution value is $F^* = \min_{0 \leq k \leq n-l, 0 \leq l \leq n} F_1(k, l)$.

PROPOSITION 4 *The DP algorithm (4) supplies an optimal solution for the $Fm|nwt, sorj| \sum C_j + \sum e_j$ problem in $O(n^3)$ time.*

Proof of Proposition 4 The optimality of (4) follows from the observation that the leftmost scheduled job j contributes its total processing time $\sum_{i=1}^m p_{ij}$ to its own completion time and an amount equal to $d'_{jl'}$ to the completion times of the remaining $k-1$ scheduled jobs. When $l > j$, there are at most $O(n^3)$ states each computed in constant time; when $l = j$, there are at most $O(n^2)$ states computed in $O(n)$ time each. Therefore, the DP algorithm (4) runs in $O(n^3)$ time. ■

Proposition 4 also applies to the more specialised fully-ordered $Fm|nwt, ord| \sum C_j + \sum e_j$ problem with $M_1 \leq M_2 \leq \dots \leq M_m$. As with Proposition 2, we can exploit the duality between no-wait and no-idle flow shops to show that a solution to the above problem is also a solution to the no-idle flow shop $Fm|nid, ord| \sum C_j + \sum e_j$ problem with $M_1 \leq M_2 \leq \dots \leq M_m$ in which the objective is to minimise the sum of the total completion time and the total outsourcing cost. The optimal solution for the $Fm|nid, ord| \sum C_j + \sum e_j$ problem with $M_1 \leq M_2 \leq \dots \leq M_m$ can be determined according to Proposition 4.

6. Conclusions

We analysed no-wait flow shop scheduling problems with rejection and showed how to augment the travelling salesman problem formulation for the no-wait flow-shop minimum makespan problem to incorporate the rejection option.

We then focused on polynomially solvable cases by considering problems with ordered jobs. We showed that the $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem is solvable in $O(n^3)$ time and that two special cases are solvable in quadratic time. We then showed that the $Fm|nwt, sorj|C_{\max} + \sum e_j$ problem is solvable in $O(n^3)$ time. We also showed that by exploiting the duality between no-wait and no-idle flow shops, management can determine which operations to outsource in a no-idle flow shop with an outsourcing option.

Our DP algorithms for the no-wait flow shop problems with the makespan objective cannot be applied to the $F2|nwt|C_{\max} + \sum e_j$ problem with arbitrary job processing times. This is because the optimal sequence of the scheduled jobs in that problem is obtained by implementing the Gilmore and Gomory (1964) algorithm which is not an index-based scheduling rule. As a result, when a job is appended to or removed from a partial sequence, the optimal ordering of the remaining jobs may change.

For the same reason, our DP algorithm cannot be applied to other ‘no-wait’ flow shop problems with rejection corresponding to special cases of the $Fm|nwt|C_{\max}$ problem solvable by variants of the Gilmore-Gomory algorithm. One such case is when all jobs have the same processing time on all but two machines and these two machines are either consecutive or one of them is the first (last) machine (van der Veen and van Dal 1991).

Our results for the two-stage no-wait flow shop problems with the minimum makespan objective also apply to the corresponding flow shop problems with blocking where there is no buffer space between the machines and any job completing its processing on the first machine must remain on it until the second machine becomes available. This is because, any two-machine schedule with blocking can be converted to a no-wait schedule with the same makespan value by shifting appropriately any blocked operation so that its completion time coincides with the start time of the other operation of the same job. The two-stage makespan equivalence also extends to flow shops with synchronous job transfers between stages analyzed by Soylu, Kirca, and Azizoglu (2007) and Waldherr and Knust (2015).

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